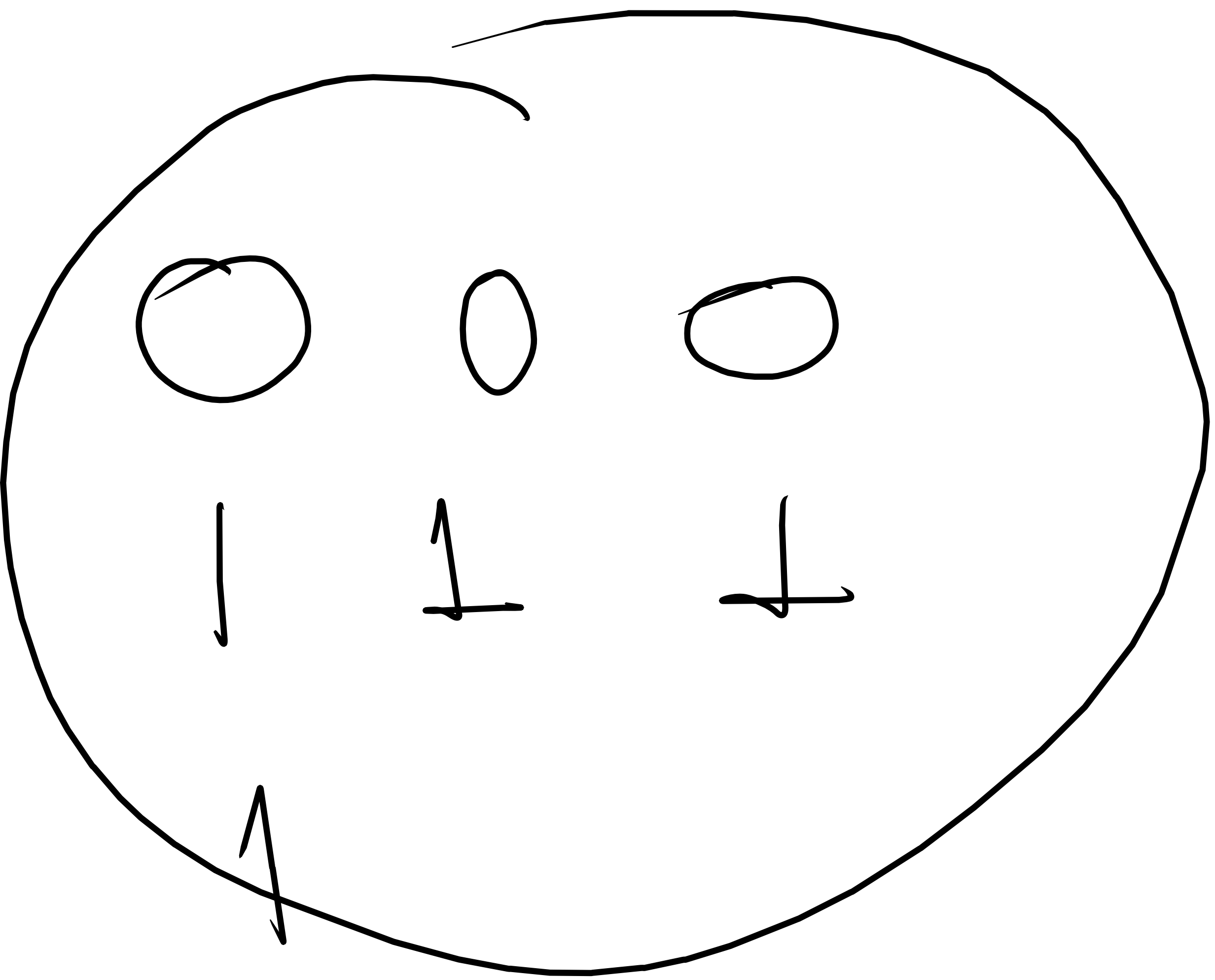
Agenda

→ Lab on CNN

→ RNN



RNN

Recurrent Neural network

Chatgpt → Remember your search / question

→ A Recurrent Neural N/w is a neural N/w designed for sequential data.

→ Unlike traditional neural network, RNN remembers previous Information.

1. Sentences
2. Speech
3. Stock price
4. weather forecast

Key Ideas :- RNN has memory

past information
↓
current Input
↓
output

Chatgpt
↓
What do think i like to learn the most ?

AI

How RNN Actualy works (step by step)

Think of an RNN as a person reading a sentence one word at a time while remembering what was read before

Sentence : I love Deep learning

An RNN sees :

Current word
+
previous memory
=
New memory

Step 1 → first word arrives

Input : I

Initialy memory is empty.

Memory = 0

RNN process ⇒ I + Memory(0)

Produces → Hidden layer h1

$$I \downarrow RNN \downarrow h1$$

Step2 : Second word arrives

Input : love

Now RNN uses :

current word = love

Previous memory = $h1$

love + $h1$ $\to$ I

love

$\downarrow$

$\boxed{\text{RNN}}$

$\downarrow$

$h1$

$\downarrow$

$h1 = I$

$h2 = I \text{ love}$

Step3 → Third word arrives

Deeplearning

$\Rightarrow$ h2 h3

Deeplearning + h1

I love Deeplearning

I love ⟶ ?

I → h1

love → h2

love + h1 → Deeplearning

I love Deeplearning

Backpropogation through time (BPTT)

error moves backwards :

$t4 \leftarrow t3 \leftarrow t2 \leftarrow t1$

weight are updated

Prediction is wrong
↓
Error
↓
updat weight

BPTT

[15, 42, 8, 9]

Movie Review website

Rottentomato

IMDb.

User reviews

15 42 8 9

This movie is fantastic → +ve

fantastic acting & story → +ve

worst movie ever → -ve

waste of money & time → -ve

Step1 → User Enters Review

T h i s

m o v i e

i s

user writes : This movie is amazing

Step2 : Tokenization

Token1 Token2 Token3 token4

convert sentence into words : ["This", "movie", "is", "amazing"]

convert to numbers : [15, 42, 8, 9]

Step3 → Embedding layer

Each word becomes vector

15 → [0, 2, 0, 7, 0, 1]

42 → [0, 5, 0, 3, 0, 9]

Input: I like this

output: posit

good → [0.8, 0.7, 0.9]

excellent → [0.9, 0.8, 0.9]

amazing → [0.85, 0.75, 0.85]

fantastic → [0.95, 0.85, 0.95]

Negative words

bad

worst

terrible

Suppose

movie = 42

amazing = 99

good = 32

bad = 5

Does no. 99 tells us "amazing" is a positive word

Embedding converts each word ID into a vector (a list of numbers)

[1, 2, 3, 4]

we get: this movie is great

[

1 → [0.2, 0.8, 0.4],

2 → [0.1, 0.3, 0.9],

3 → [0.5, 0.6, 0.2],

4 → [0.9, 0.8, 0.7]

]

bad = [-0.8, -0.7, -0.9]

worst → [-0.9, -0.8, -0.5]

Step4 RNN Reads one word at a time.

This

h1

movie

movie + h1 = h2

This movie is

h2 + is = h3

Dense layer

final memory

positive

98%

negative

2%